

JUNGWOO SON

GRADUATE AUTOMOTIVE ENGINEER

(+44) 07442 657340 | jungwoo1731@gmail.com | www.linkedin.com/in/json1731

PROFILE

MEng Automotive Engineering student with industry experience in structural simulation, crash analysis and composite modelling. Experienced in applying FEA to support design decisions under real-world constraints, with a strong interest in high-performance and safety-critical engineering.

EDUCATION

MEng Automotive Engineering

Loughborough University, UK | SEP 2019 – JUN 2026

Grades: Year 2 – 72.1%, Year 3 – 72.7% | Projected First Class

Relevant modules: Finite Element Methods, Stress and Structural Analysis, Crashworthiness, Composite Materials, Vehicle Dynamics and Simulation

PROFESSIONAL EXPERIENCE

CAE Engineer

TriMech Simulation Solutions, UK | JUL 2023 – AUG 2024

- Supported automotive clients in structural validation and optimisation under time and delivery constraints
- Performed crash and occupant safety simulations using LS-DYNA and ANSA to evaluate structural response
- Developed a composite door structure meeting mass and side-impact requirements through iterative optimisation
- Created and validated a composite material model used across crash and NVH analyses
- Presented simulation results and recommendations in client design reviews

PROJECT EXPERIENCE

Concept Rescue Vehicle Design

Team Leader & Chassis Design | Loughborough University | SEP 2024 – APR 2025

- Led a 6-member multidisciplinary team, coordinating subsystem integration and project delivery
- Performed FEA-based chassis optimisation to balance stiffness and mass
- Delivered a final technical output integrating engineering validation with commercial feasibility

Formula Student Composite Chassis Concept

Team Leader | Loughborough University | OCT 2025 – APR 2026

- Led design and analysis of a carbon fibre monocoque chassis under Formula Student regulations
- Conducted carbon prepreg manufacturing and tensile testing to characterise material behaviour
- Performed correlation between experimental results and simulation models to improve analysis accuracy
- Oversaw team activities, supporting members in resolving technical challenges and maintaining project progress

Final Year Project – Crash Energy Absorber (Ongoing)

Loughborough University | OCT 2025 – Present

- Designing and optimising a corrugated energy absorber to improve impact performance under quasi-static loading
- Investigating the influence of corrugation geometry on deformation modes and energy absorption behaviour
- Using non-linear analysis in MSC Marc to evaluate structural response and guide design refinement

ADDITIONAL EXPERIENCE

Republic of Korea Army

Military Service | DEC 2020 – JUN 2022

- Completed mandatory military service in a structured and high-responsibility environment
- Served as a squad leader, responsible for coordinating personnel and maintaining team discipline
- Supported administrative and operational tasks, ensuring accurate documentation and adherence to procedures

SKILLS

Simulation & Analysis

LS-DYNA, MSC Marc, Genesis

Pre/Post Processing & CAD

ANSA, Siemens NX

Programming & Modelling

MATLAB, Simulink

Engineering Areas

FEA, Crashworthiness, Composite Materials, Vehicle Dynamics

Core Competencies

Analytical problem solving, Technical communication, Project coordination, Cross-functional teamwork

ADDITIONAL INFORMATION

Languages

English (Fluent), Korean (Native), German (Beginner)

Qualifications

Full UK driving licence